Kwanho Kim

SOFTWARE ENGINEER
C++ DEVICE SOFTWARE & INTEGRATION

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C++ engineer building device software for semiconductor inspection equipment at KLA, after robot-vision integration at Pickit 3D and V2X prototyping at Bosch. Focus: serial device control, fault recovery, and logging that makes field failures diagnosable.

EXPERIENCE

KLA C++ Engineer — Aug 2026 – Present · Leuven, Belgium

- Designing and implementing fault recovery for the equipment's lift mechanism in C++ (in progress)
- Exposed the camera's device-reported **white-balance settings** in the operator UI.
- Refactored **serial device control and logging** into a shared equipment layer.

Pickit 3D Research & Software Engineer — Aug 2025 – Mar 2026 · Leuven, Belgium

- Traced a **field communication failure** to a port conflict, fixed it and documented recurrence checks.
- Built a timezone-aware **log download and conversion workflow** with a multilingual UI for triage.
- Built a reusable **Python WebSocket client** with reconnect, backoff and message validation.
- Developed system and network **configuration screens** with input validation and safe defaults.

Bosch V2X Communication Intern — Jun 2024 – Nov 2024 · Hildesheim, Germany

- Built a **V2X demonstrator** for safety-critical scenarios; validated end-to-end messaging in test runs.
- Integrated communication modules behind explicit **data contracts** between components.
- Built the **operator interface** for an eBike safety prototype in React.

PROJECTS

WINNER **1st** — **Color Saves Life** — Feb 2024
digital.auto & mobility challenge · BCX 2024
Built the Python client for a driver visual-assistance prototype, integrated eye tracking with simulation and presented the team's demo.

TECHNICAL SKILLS

Languages	C++17 (primary; studying C++23/26), C, Python; TypeScript / JavaScript, Bash
Device software	Serial device control, logging and diagnostics; fault-recovery design; CAN (basic)
Integration	REST, WebSocket, D-Bus, ROS 1/2; Docker; CARLA simulation
Operator UIs	Qt/QML, React
Tooling & AI	Git, Make; AI coding agents (Claude Code, Devin) for refactoring and tests

EDUCATION

Mechatronics · Bachelor's in progress · KNOU Prime College — Feb 2026 – Present
Artificial Intelligence · Bachelor's · Korea National Open University — Mar 2023 – Feb 2025
Business Administration · Bachelor's · Academic Credit Bank System — Apr 2019 – Aug 2022
Computer Engineering · Bachelor's · Academic Credit Bank System — Apr 2019 – Aug 2020

TRAINING

42 · Software engineering · Seoul / Wolfsburg — Completed
SEA:ME · Automotive & Mobility · Wolfsburg — Jul 2023 – Jun 2024

Korean: Native · English: Fluent (IELTS 7.5, 2025) · Japanese: Learner